

Project Concept Note

FitCoach AI — a custom GPT-style fitness coaching assistant

Project Title and Application Name

FitCoach AI

Problem Statement / Objective

Many people want a knowledgeable, encouraging fitness coach available on demand, without the cost or scheduling friction of a personal trainer. FitCoach AI provides an always-available conversational assistant that helps users plan workouts, understand proper exercise form, and stay motivated, while clearly deferring to qualified professionals for medical or injury-related concerns.

Target User and Use Case

- Beginners who want a structured starting point for training without a gym membership advisor.
- Intermediate lifters or runners who want quick answers on programming, form cues, or recovery.
- Anyone who wants a motivational check-in to stay consistent with a fitness routine.

LLM Model and API Used

Anthropic Claude, accessed through the Anthropic Messages API (streaming mode), called from a Python FastAPI backend. The API key is stored server-side as an environment variable and is never exposed to the frontend or committed to version control.

Key Features of the Application

- Custom GPT-style persona: FitCoach AI, a fitness-focused system prompt that scopes the assistant to training, form, and motivation topics.
- Responsive frontend (HTML/CSS/JavaScript) that works on both desktop and mobile.
- FastAPI backend that routes requests to the Anthropic API and keeps the API key secret server-side.
- Streaming responses: text renders progressively in the chat UI rather than appearing all at once.
- Containerized with Docker for consistent, portable deployment.
- Deployed to AWS (App Runner) behind a public HTTPS URL.
- Security best practices: no secrets in frontend code or git history; credentials injected via environment variables / AWS Secrets Manager.

Expected User Experience and Outcomes

A user opens the web app on any device, types a fitness question or goal into the chat box, and sees FitCoach AI's response stream in token-by-token in real time, similar to using a modern AI chat product. The tone is encouraging and practical. For anything outside safe scope (diagnosing pain, prescribing medical nutrition plans), the assistant redirects the user to a qualified professional rather than guessing. The outcome is a lightweight, always-on coaching companion that lowers the barrier to starting and sticking with a fitness routine.

Live Application URL

*[Paste your AWS App Runner HTTPS URL here after deployment, e.g.
https://xxxxxxx.us-east-1.awsapprunner.com]*